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Exposure of *Cyprinus carpio* var. larvae to PVC microplastics reveals significant immunological alterations and irreversible histological organ damage

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ABSTRACT

Microplastics have become pervasive in ambient aquatic ecosystems over the last decade and are currently a serious global scale concern. To evaluate the potential toxic effects of PVC (polyvinyl chloride) microplastics on the immune functions of freshwater fish, this study undertook a conducted chronic 60-day dietary exposure experiment with *Cyprinus carpio* var. larvae. We exposed the fish to four microplastic treatments of different concentrations (food rationed diets): no-plastic (control), 10%, 20% and 30%. At the end of the experimental period the impacts of microplastics on the histology, biochemistry, ROS (reactive oxygen species) levels and gene transcription of immune organs were investigated. The results revealed that PVC microplastics induced cytoplasmic vacuolation in the liver, damaged villi in the intestine, inflammatory cell infiltration, hemosiderosis and vacuolar degeneration in the spleen, glomeruli tuft shrinkage and aggregation of melanin macrophage cells in the kidney. Moreover, following PVC microplastics exposure, ROS levels in the liver and protein levels of pro-inflammatory cytokines including IL-6, IL-8, and TNFα in the liver and serum were increased. Furthermore, modifications in the activities of non-specific immunoenzyme ACP (Acid phosphatase), AKP (alkaline phosphatase), LZM (lysozyme), and expression levels of a range of immune-related genes were observed. Using various techniques at the histological, biochemical and molecular levels, our findings demonstrated the effects of PVC microplastics on changes and imbalances in the immune status of carp. The results of this study provide basic toxicological data toward elucidating and quantifying the impacts of microplastics immunotoxicity on aquatic organisms.

1. Introduction

The significantly increased anthropogenic use of non-degradable plastic products has resulted in their ubiquitous accumulation on a global scale. Due to the chemical stability and slow degradation of plastic products, the accretion of small plastic fragments in ambient aqueous environments is particularly obvious (Jambeck et al., 2015). Among them, plastic fragments having diameters of < 5 mm are internationally considered as microplastics (Barboza and Gimenez, 2015; Koehler et al., 2015). Recently, increasing attention has been focused on the adverse impacts of microplastics on aquatic ecosystem. Studies have revealed that polystyrene microplastics can induce microbiota imbalances and inflammation in the guts of adult zebrafish (Jin et al., 2018), which leads to apoptosis and an increase in necrotic red blood cells (Atbga et al., 2020). Exposure to various concentrations of polymeric microplastics for 30 days can cause sublethal effects on the growth and behavior of *Japanese medaka* larvae (Pannetier et al., 2019). In zebrafish (*Danio rerio*), inflammation, hepatic steatosis and oxidative stress were induced after treatment of PS microspheres for three weeks (Lu et al., 2016).

As species at the top of the aquatic food chain, fish are more vulnerable when toxins infiltrate the nutrient web via bio-magnifications; thus, they are extensively used as bio-indicators of environmental levels (Abdel-Moneim et al., 2012; Moyson et al., 2016). Although progressively more studies have verified the potential deleterious effects of microplastics intake on marine life, only few have investigated the potential immune impacts of microplastics on freshwater ecosystems. In general, the immune systems of fish are very sensitive to environmental or internal stresses. One factor is that the immune organs of fish, as the primary barriers between the external

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environment and closely regulated internal milieu, play a critical role in protecting fish against infection (Grinwis et al., 2009; Anderson and Zeeman, 1995). Additionally, fish respond quickly to poisons in aqueous environments by altering the activities or contents of immune parameters, including acid phosphatase (ACP), lysozyme (LZM) and cytokines (Ma and Li, 2015). It is fascinating and meaningful to investigate the effect and potential mechanisms through which microplastics initiate changes or/and imbalances in the immune status of the above mentioned tissues and processes.

Yellow River carp (*Cyprinus carpio* var.), with a long history of breeding in China, are commonly selected as experimental models and exposed to environmental contaminations to evaluate the health of aquatic ecosystems (Chen et al., 2015). The present study explored the potentiality of PVC microplastics in *Cyprinus carpio* var., i.e. how it influences the growth and microstructure of immune organs, non-specific immune enzymes and pro-inflammatory cytokines, immune-related genes expression and ROS levels.

2. Materials and methods

2.1. Chemicals, animals and treatment

The average diameters of PVC microplastics (Sigma-Aldrich, Lot# MKBW2191V) range from 100 to 200 μm. The *Cyprinus carpio* var. larvae used for this study were in the 115–125-day groups and obtained from the aquaculture center of Henan Normal University. The mean body weights and lengths of the fish were 11.76 ± 1.51 g and 7.38 ± 0.66 cm, respectively.

The animals were randomly divided into four groups (n = 50), including a control group (no PVC microplastics) and three experimental groups (10%, 20%, and 30% microplastics by weight, respectively). All fish were treated by dietary exposure for 60 days. The exposure concentrations were determined based on environmentally relevant concentrations found in surface freshwater ecosystems (Oliveira et al., 2020; Horton et al., 2017). The performed fish maintenance and diet preparation are described in Supplemental Material and a previous study (Xia et al., 2020).

2.2. Growth performance

Every six days, the fish (10 larvae per group) were measured for weight and length. After 60 days, the fish were measured following anaesthetization with tricaine methane sulfonate, and then prepared for further studies.

2.3. Histological examination

The livers, intestines, spleens and kidneys of six fish were randomly sampled from each tank. After 24 h of formalin fixation, the tissues were dehydrated by graded ethanol and xylene, and embedded in paraffin. Subsequently, they were sectioned at 6 μm and stained with hematoxylin and eosin (H&E).

2.4. Real-time quantitative PCR

At the end of exposure treatments the gene transcription levels were determined as previously described (Xia et al., 2020), using 20 mg of each tissue from three fish. The total RNA was extracted from livers, intestines and spleens using TRIzol (RNA Extraction Kit, Invitrogen, USA) and converted to cDNA using Prime Script reverse transcriptase (Takara, Japan). The mRNA expressions were measured via a three-step SYBR® QRT-PCR kit (Takara, China) using the QT system (Thermo, USA). All samples were analyzed in triplicate, and the reactions were independently repeated twice to ensure reproducibility. The elongation factor 1 alpha (*elfα*) gene was used as a reference to normalize cDNA loading (Table S1).

2.5. Non-specific immunoenzyme analysis

At the end of the 60-day experimental periods, the livers, intestines, and gills extracted from 12 fish in each experimental unit were utilized to measure the non-specific activities of immunoenzymes. The activities of ACP, LZM and alkaline phosphatase (AKP) were assayed using a commercial kit (Jiancheng Bioengineering Institute, Nanjing, China) according to the manufacturer's instructions. One king unit (1 king unit = 7.14 U/L) of ACP activity was represented per gram of tissue resident protein, producing 1 mg phenol at 37 °C for 30 min monitored at 520 nm. One king unit of AKP activity was represented per gram of tissue resident protein, in tissue generating 1 mg phenol at 37 °C for 15 min monitored at 520 nm.

2.6. Pro-inflammatory cytokine analysis/ ELISA assays

Following 60-days of PVC microplastic exposure, the livers and caudal vein blood samples of the carp larvae were collected for the distinct quantitative detection of pro-inflammatory cytokines. The samples and standards were prepared in triplicate, and the IL-6, IL-8 and TNFα contents were evaluated using fish specific ELISA (enzyme-linked immune sorbent assay, ELISA) kits (MyBioSource, San Diego, USA) according to the manufacturer's instructions. Briefly, 10 μL volumes of serum were added to 96-well microtiter plates coated with specific antibodies, to which 40 μL of the sample diluent and 50 μL of horseradish peroxidase-labeled antibodies were added. The plate was incubated at 37 °C for 1 h and then rinsed five times with a buffer. Next, added 50 μL of hydrogen peroxide and tetramethyl benzidine were added, respectively, and the plates were incubated at 37 °C for 15 min. Finally, 50 μL of vitriol was added to stop the reaction, and the optical density (OD) values were determined at 450 nm.

2.7. ROS detection

To analyze liver oxidative stress levels, the production of ROS (reactive oxygen species, ROS) was detected by DCFH-DA (2′,7′-Dichlorodihydrofluorescein diacetate), which transformed into highly fluorescent DCF when the cells were oxidized. Specifically, the fluorescence intensity of DCF was proportional to the level of cellular ROS. Following 60-days of exposure, the hepatocytes of three fish were isolated and suspended using the ROS assay kit (CA1410, solarbio) according to the manufacturer's instructions. The hepatocytes were incubated with 10 μm DCFH-DA diluted in PBS at 37 ℃ for 1 h. After rinsing twice with PBS, the cells were suspended in PBS. Subsequently, 10,000 cell specific events were evaluated by flow cytometry (Becton Dickinson, FACSVerse) and Flowjo 7.6 software was used to analyze the fluorescence intensity.

2.8. Data analysis

Statistical analysis was performed using one-way analysis of variance (ANOVA) and SPSS 13.0 software (SPSS Inc.). The differences between the treatments were determined by the least significant difference (LSD) test, where probability levels of $p < 0.05$ were considered as statistically significant.

3. Results

3.1. Carp growth performance following exposure

Although no fish mortality occurred within the 60 days of exposure, scale abscission was observed in the 20% (15/50) and 30% PVC (23/50) concentration groups (Fig. 1B,C). Additionally, in the 30% PVC concentration groups, the microplastics diets led to obvious skin hemorrhaging (9/50). As indicated in Fig. 1D, the weight gain and body lengths of the fish in all the exposure groups were significantly reduced

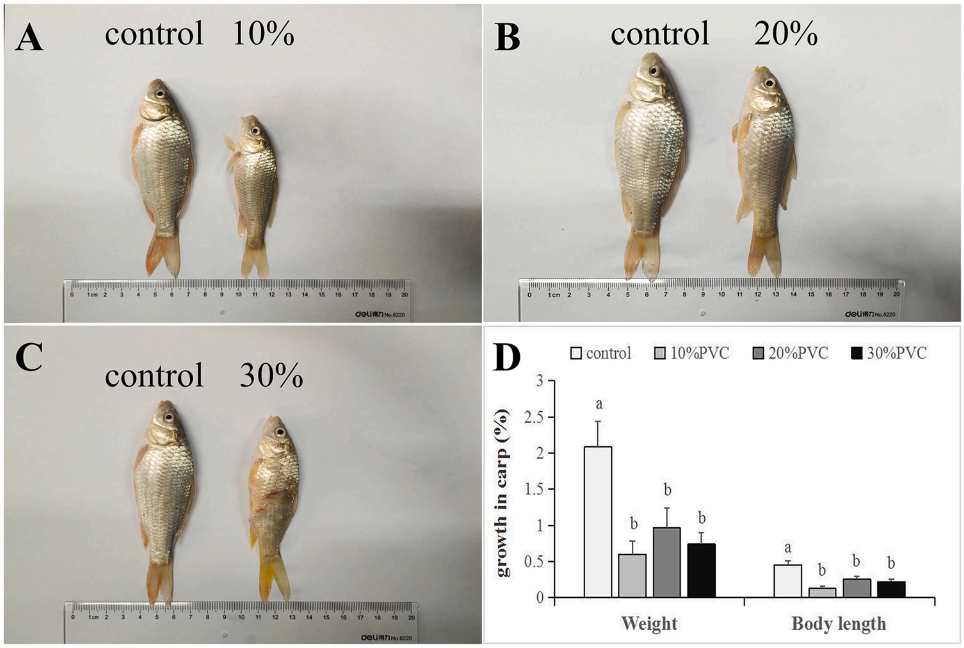

Fig. 1. Growth performance of *Cyprinus carpio* var. after 60 days of exposure to PVC microplastics (A-C); Growth percentage of *Cyprinus carpio* var. after 60 days of exposure. Error bars are used to represent standard deviation (D). Significant differences ($P < 0.05$) between treatments and controls are indicated by the different lowercase letters above the bars.

in contrast to the control group, particularly those in the 10% treatment group.

3.2. Histological changes induced by microplastics treatments

Compared with the control group, the livers, intestines, spleens, and kidneys of the carps exposed to PVC were damaged to varying degrees. As displayed in Fig. 2, the livers of the PVC treatment groups cells became obviously loosely arranged, while in the control group hepatocytes showed a normal radial arrangement. Moreover, cytoplasmic vacuolation (Fig. 2B-D[black arrows]) were found in the livers of the fish in all PVC concentration groups. In the gut, some of the intestinal villus were destroyed (Fig. 2F-H[black arrows]) and more vacuolation appeared (Fig. 2F′,G′[blue arrows]). In the spleen, hemosiderosis (Fig. 2J-L[green arrows]) and vacuolar degeneration (Fig. 2K,L[black arrows]) was apparent. Further, in the kidneys, melanin macrophage accumulation was observed (Fig. 2N-P[black arrows]), the glomeruli tufts began to contract (Fig. 2N-P[blue arrows]), and renal tubules became denatured (Fig. 2N-P[green arrows] and N′).

3.3. Expression of mRNA genes involved in immune processes

We examined changes in the expressions of *IL-1β*, *IFNγ*, *TNFα*, *IL-6*, *IL-8*, *gcC3*, and *LZM* in response to 60 days of PVC microplastics exposure. As displayed in Fig. 3, for the 10% and 20% PVC concentration groups, the expressions of *IFNγ* and *TNFα* in the liver were significantly down-regulated, whereas those of *IL-1β*, *IL-6* and *gcC3* were significantly up-regulated. Additionally, the expression of immune-related genes in the intestine for the 10% PVC concentration groups were significantly increased compared with the control group. Conversely, in the spleen,

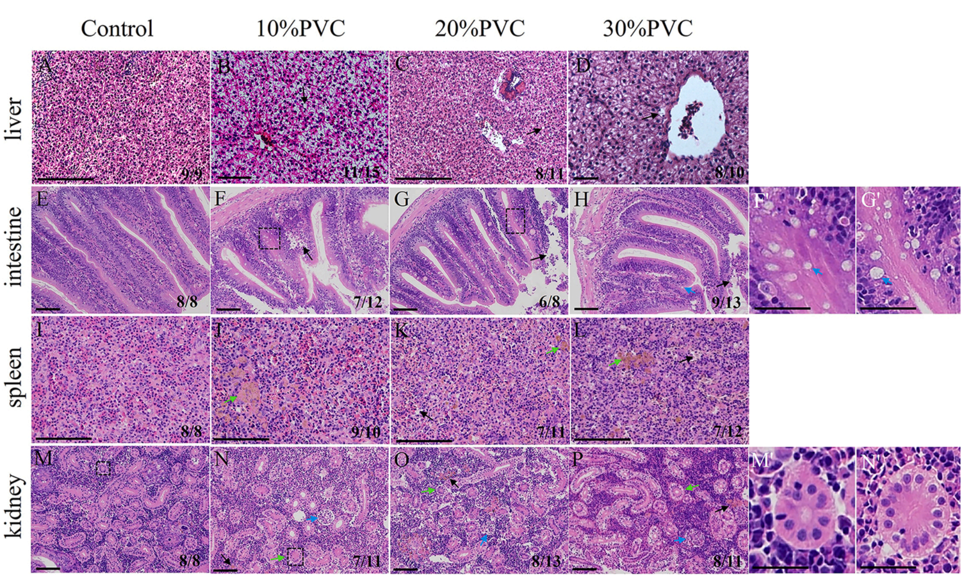

Fig. 2. Histological analysis of the longitudinal section of liver (A-D), intestine (E-H), spleen (I-L) and kidney (M-P) after 60-day PVC microplastics exposure in *Cyprinus carpio* var. F′, G′, M′, N′, Magnification of the boxed area in F, G, M, N. Black arrows (B-D) show cytoplasmic vacuolation in the liver. Black arrows (F-H) show damaged villi in the intestine. Blue arrows (H,F′,G′) show vacuoles in swollen villi. Green arrows (J-L) and black arrows (K,L) show hemosiderosis and vacuolar degeneration in the spleen, respectively. Blue arrows (N-P), black arrows (N-P) and green arrows (N-P) show shrinkage of glomeruli tuft, aggregation of melanin macrophage cells and tubular degeneration in the kidney, respectively. Numbers in each panel represent the number of sections with the pattern visible over all the sections that were studied. Scale bar, 50 μm.

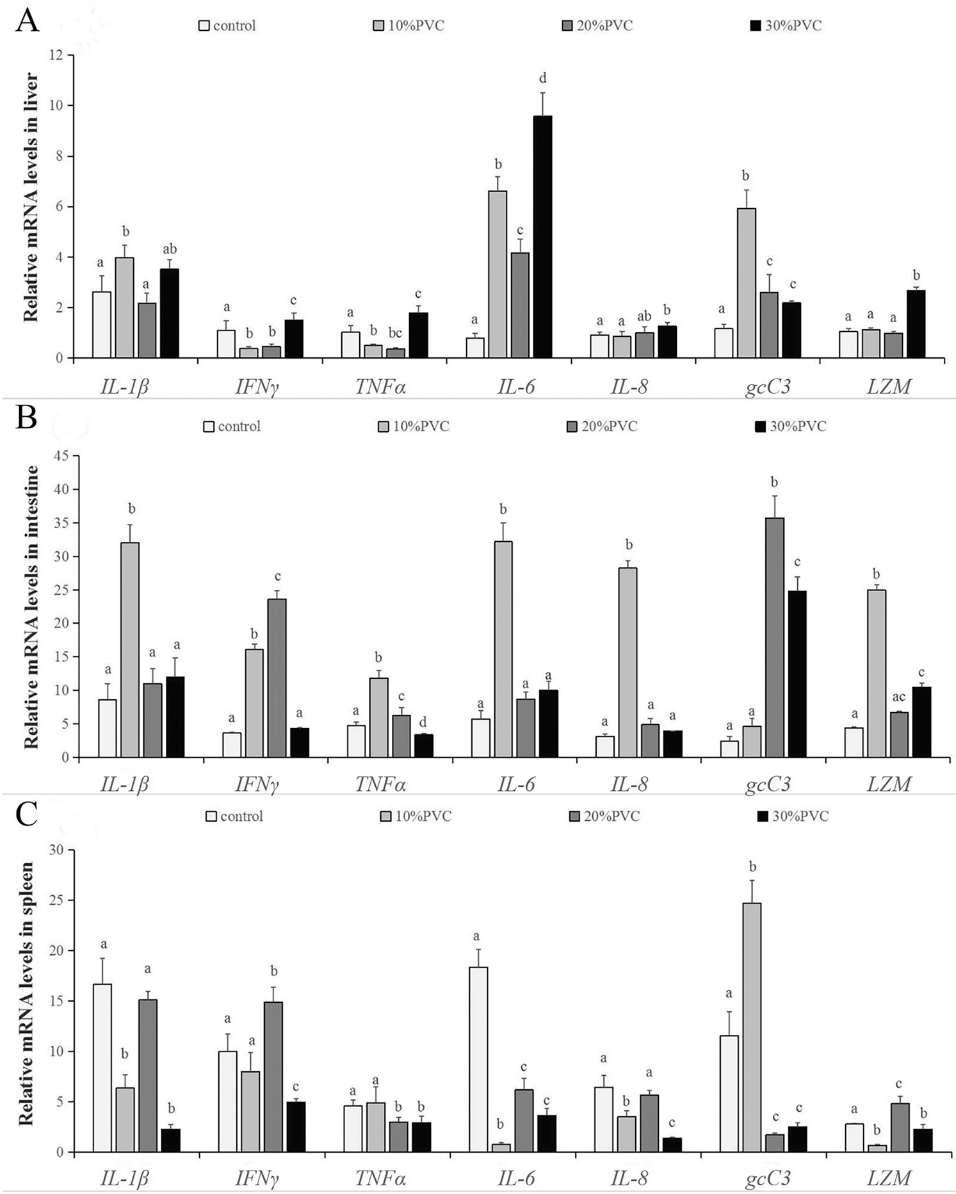

Fig. 3. The transcriptional changes of immune-related genes including *IL-1β*, *IFNγ*, *TNFα*, *IL-6*, *IL-8*, *gcC3* and *LZM* in the liver (A), intestine (B) and spleen (C) of *Cyprinus carpio* var. after 60 days of PVC microplastics exposure. A column represents the average result of trials conducted in triplicate. The standard deviation is represented by error bars. The different lowercase letters represent the significant differences ($p < 0.05$) between the treatments and controls.

except for the up-regulated expression of *gcC3* in the 10% PVC concentration group and *IFNγ* and *LZM* in the 20% PVC concentration groups, the expressions of other immune-related genes were significantly down-regulated in the treatment groups.

3.4. Effects of PVC microplastics on activities of enzyme related non-specific immunity in carp

The results indicated that the activities of ACP and AKP in the fish livers and intestines were significantly down-regulated compared with the control group. However, different results were observed in the gills, where the activities of ACP and AKP were significantly increased in contrast to the control. Moreover, their activities in the gills gradually decreased as the PVC concentrations increased (Fig. 4A, B). In livers and intestines, the activities of LZM were significantly up-regulated compared with the control groups (Fig. 4C).

3.5. Effects of PVC microplastics on the secretion of pro-inflammatory cytokines

As shown in Fig. 5, the results revealed that the levels of IL-6, IL-8 and TNFα in the livers under all treatment groups were significantly enhanced in contrast to the control group. As to the IL-6 levels in serum, it initially ascended and then descended to a level that did not differ significantly from the control. The levels of TNFα and IL-8 in serum were up-regulated in the 10% PVC concentration groups. Although they were slightly down-regulated in the 20% PVC concentration group, the levels remained significantlyup-regulated in the 30% PVC concentration group in contrast to the control.

3.6. Microplastics exposure induced ROS production in carp hepatocytes

DCFH can be oxidized to DCF with high fluorescence by intracellular ROS. As shown in Fig. 6A, the peak areas of different colors represent the population of DCF positive cells in different groups. According to flow

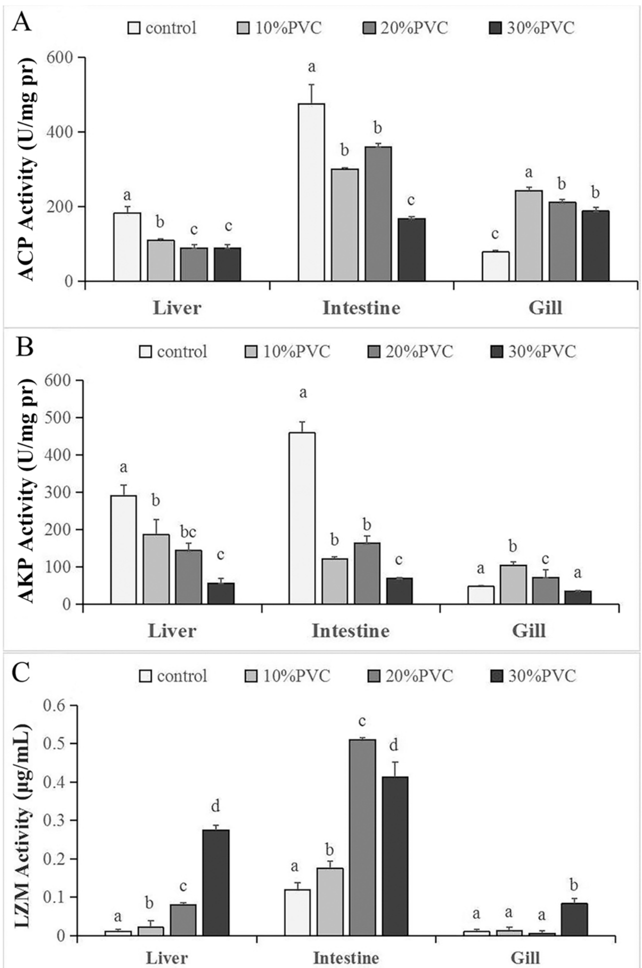

Fig. 4. The activities of acid phosphatase (ACP) (A), alkaline phosphatase (AKP) (B) and lysozymes (LZM) (C) in the liver, intestine, and gills of *Cyprinus carpio* var. after 60 days of PVC microplastics exposure. A column represents the average result of trials conducted in triplicate. The standard deviation is represented by error bars. The different lowercase letters represent the significant differences ($p < 0.05$) between the treatments and controls. Pr, protein.

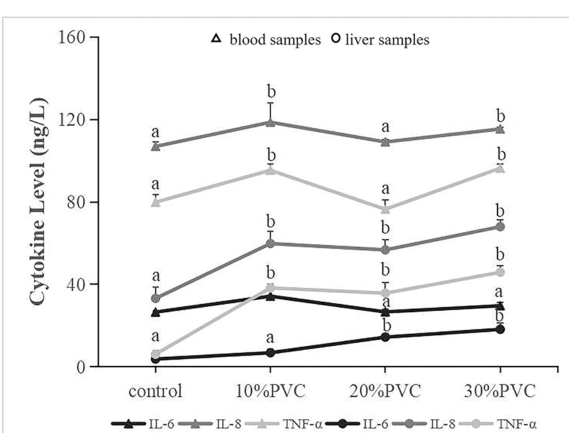

Fig. 5. The level of immunocytokine (IL-6, IL-8 and TNFα) in the liver and blood of *Cyprinus carpio* var. after 60 days of PVC microplastics exposure. A column represents the average result of trials conducted in triplicate. The standard deviation is represented by error bars. The different lowercase letters represent the significant differences ($p < 0.05$) between the treatments and controls.

cytometry analysis, compared with the control group, the fluorescence intensity increased slightly in the 10% PVC treatment groups (Fig. 6B). Under the 20% and 30% PVC treatment groups, the fluorescence intensity increased sharply in contrast to the control group, indicating a remarkably high intracellular ROS levels. These results exhibited a distinct dose dependent uptrend following 60 days of exposure.

4. Discussion

Currently, microplastics permeate the ambient atmospheric, soil, and aqueous environments and seriously exacerbate health risks for living organisms (Khalid et al., 2021; Mammo et al., 2020). This study exposed Yellow River carp larvae to different concentrations of PVC microplastics to investigate their detrimental impacts on organs, immune parameters, immune related gene expression and ROS levels.

4.1. Histological changes

Histopathologic changes are vital indicators for the detection of exposure to environmental stressors in fish (Wang et al., 2013). The liver is the primary organ that performs myriad key metabolic pathways; thus, the toxic effects of chemicals generally occur in the liver (Yildirim et al., 2010). Microplastics exposure has been reported to induce inflammation and fat accumulation in Nile Tilapia and Chinese mitten crab, which ultimately led to liver damage (Rochman et al., 2013; Lu et al., 2018). Similarly, we observed fatty vacuolation in the livers of fish in the PVC treatment group. The formation of fat vacuoles primarily affects the balance of fat metabolism in hepatocytes via xenobiotics (Wester and Canton, 1987).

The accumulation of harmful substances can cause pathological damage to intestinal tissues, the morphology of which may be used as an important index to evaluate the impacts of environmental pollutants (Tashjian et al., 2006). Following PVC exposure, we observed villi destruction, including the partial rupture of intestinal villus, intestinal villus swelling and vacuolation, while the control gut microvilli remained intact. On one hand, the integrity of intestinal tissue is critical for maintaining normal intestinal barrier function and the capacity for digestion and absorption in fish (Gao et al., 2013). On the other hand, changes in intestinal morphologies are typically correlated with intestinal dysfunction, which inhibits the growth of fish (Kang et al., 2021; Duan et al., 2020). This was confirmed by our results, which revealed reductions in body weights and decreased lengths in carp larvae.

As far as is known, the spleen is the only lymph node organ of bony fish (Nemcsók et al., 1984) that contains abundant lymphocytes and macrophages. Macrophages can engulf heterogeneous particles such as microorganisms, macromolecules, as well as injured or apoptotic tissues (Xia and Triffitt, 2006). They can also provide a protective inflammatory response and rebuild tissue integrity through the production of effector molecules (Atri et al., 2018). As our results indicated, the aggregation of macrophages occurred in both the spleen and kidneys. This suggested that PVC microplastics induced elevated levels of oxidative stress in these immune organs and initiated changes in immune status through the production of pro-inflammatory chemokines and cytokines, recognizing and eliminating foreign antigens via phagocytosis.

4.2. Gene expression

IL-1β is a distinctly essential inflammatory cytokine, which plays a pivotal role during the initial phase of inflammation (Groslambert and Py, 2018). Further, IL-1β and TNFα are critical inducers of IL-8, which are important for the activation of neutrophils and macrophages at infection sites (Wei et al., 2018). In the present study, it was worthy of note that most immune-related genes (*IL-1β*, INF *γ*, *TNFα*, *IL-8*, *gcC3*, and *LZM*) exhibited remarkably higher expressions in the intestine than the liver and spleen. This might be explained by the fact that the intestine is a core site for fermentation, digestion, and the absorption of nutrients

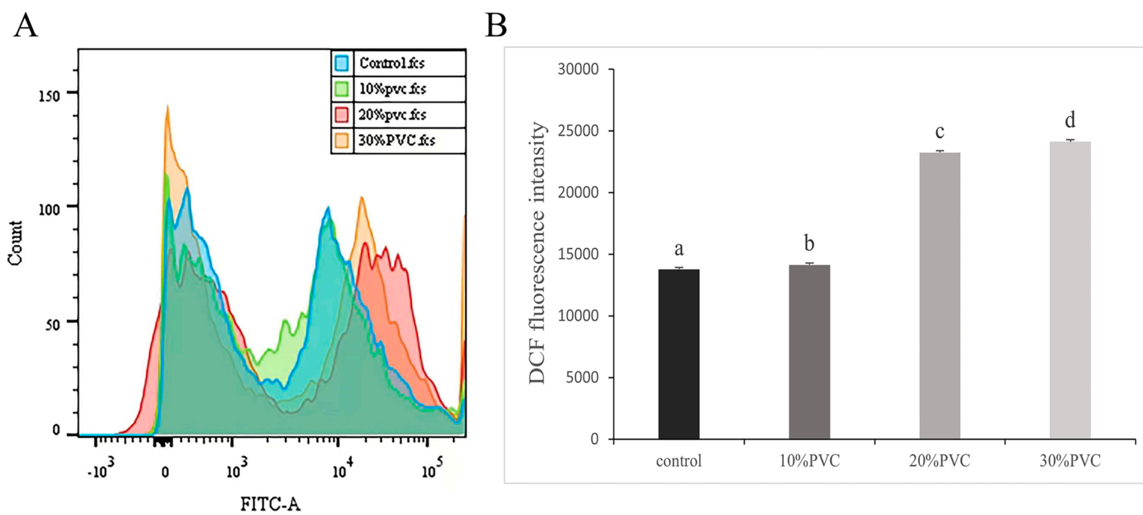

Fig. 6. Histogram analysis (A) and DCF fluorescence intensity of ROS level detected by flow cytometry in hepatocytes (B). DCF, dichlorofluorescein. A column represents the average result of trials conducted in triplicate. The standard deviation is represented by error bars. The different lowercase letters represent the significant differences ($p < 0.05$) between the treatments and controls.

(Gonçalves et al., 2007). Furthermore, once food passes through the digestive tract, it remains in the intestine for a prolonged duration. During this time the food might carry foreign bacteria that generate an immune response.

Moreover, the *gcC3* levels in the intestines of the 20%, 30% PVC concentration groups and in the livers, spleens of the 10% PVC concentration group were extremely high in contrast to the control groups. This may have been due to the fact that complement 3 is a pivotal composition of the innate immune systems of fish, which is extraordinary vital for the linkage between innate and adaptive immunity (Ichiki et al., 2012). Several previous reports have demonstrated that environmental toxins affect C3 expression levels and disrupt the innate immune systems of fish (Das et al., 2011). The increase in C3 was also observed in the livers of redfin puffer fish that were exposed to tetrodotoxin (Matsumoto et al., 2011).

4.3. Immune function

Compared with other higher vertebrates, nonspecific immunity in teleost fish is prominent for disease resistance, in which LZM, AKP, and ACP play key roles (Iwasaki and Medzhitov, 2010). Jia et al. (2020) suggested that long-term exposure to hydrogen peroxide induced a significant increase in the activities of ACP and AKP in the livers, muscles, and intestines of carp, while reducing them in the gills, which was consistent with our results. Likewise, an earlier study showed that a QBA treatment could severely inhibit the activities of ACP and AKP in the livers of Oncomelania snails (Ke et al., 2019). It was reported that the decrease of hemocyte counts hindered the synthesis of immune enzymes (Hong et al., 2020). Further, another study indicated that high salt levels increased the activities of ACP and AKP in the gills of freshwater mussels (Zhang et al., 2021). LZM is a cationic enzyme that attacks the peptidoglycan in bacterial cell walls and serves as a nonspecific innate immunity molecule against the incursion of detrimental bacteria (Luo et al., 2007). Thus, the modification of LZM activities is typically employed as an important index of innate immunity functions (Kong et al., 2012). In this study, the LZM activities in the intestine increased more significantly than that in the liver and gills. As relates to the up-regulation of the LZM expression level in intestine, the relationships between these two results were further considered.

The immune status of fish is intimately correlated with inflammation, which is initiated and regulated via inflammatory cytokines (Secombes et al., 2001; Lin et al., 2020). It has been shown that TNFα is involved in cell proliferation, inflammatory responses and the stimulation of the immune system (Old et al., 1988; Thommesen and Laegreid, 2005). As acute phase response factors and important pro-inflammatory cytokines, IL-6 and IL-8 recruit and activate macrophages and neutrophils to remove cell debris in response to tissue injuries (Slavov et al., 2005; Strowig et al., 2012). In this study, the up-regulation of cytokines associated with inflammatory disease responses was triggered at the sites of microplastics-induced organ injury (Papadakis and Targan, 2000). Changes in pro-inflammatory cytokines in the liver may be indicators of PVC microplastics-induced immunotoxicity and liver damage, as these cytokines are essential for the regulation of inflammatory processes. The cross-corroboration of these findings demonstrated that immune organs, immune components and cytokines act in unison. Once microplastics impact one of these aspects, the resulting effects may cascade into such a synergistic whole.

4.4. Oxidative stress

The body possesses effective antioxidant defense mechanisms to remove the ROS generated by animal cells under normal physiological conditions. Oxidative stress arises when there is an imbalance between ROS production and clearance (Shen et al., 2010). Fish have well developed antioxidant defense systems that are designed to neutralize the toxic effects of ROS, which may render fish susceptible to disease (Sheikhzadeh et al., 2012; Zhang et al., 2004). In our earlier studies, we observed a decrease in the activities of SOD, CAT and GPx in fish livers following PVC microplastics exposure (published). The present results indicated that the ROS levels of hepatocytes steadily increased in a dose-dependent manner, which suggested that at higher concentrations, PVC microplastics promote the generation of ROS.Interestingly, studies have confirmed that the immune system easily releases ROS when subjected to oxidative stress (Bogdan et al., 2000), as ROS may be employed as a signaling molecule to induce the generation of pro-inflammatory cytokines within immune cells (Naik and Dixit, 2011). This means that the release of ROS may induce the up-regulation of cytokines in the liver. Conversely, oxidative stress can initiate tissue damage that induces the release of pro-inflammatory cytokines into the blood (Ma et al., 2018), which may be behind the increase in serum cytokine levels.

5. Conclusion

Overall, we focused on how microplastics affected the immune-toxicological responses of Cyprinus carpio var. larvae. According to the experimental findings of this study, exposure to PVC microplastics prevented the carp from growing and induced oxidative damage to the intestine, spleen, kidney, and liver. The altered expressions of immune-related genes and the activities of non-specific immunoenzymes were observed in the immune organs of larvae exposed to PVC microplastics. Additionally, PVC microplastics increased the contents of pro-inflammatory cytokines and the ROS levels in the liver. Thus, our findings implied that PVC microplastics exposure induced changes in the immune status of fish on histological, molecular and biochemical levels. These data may provide valuable toxicological references for further research into microplastics-induced immune defenses of freshwater ecosystems.

CRediT authorship contribution statement

Xinya Liu: Data curation, Visualization, Writing – original draft, Methodology. **Chaonan Liang:** Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Miao Zhou:** Formal analysis, Investigation. **Zhongjie Chang:** Supervision. **Li Li:** Conceptualization, Funding acquisition, Resources, Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

Acknowledgments

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.ecoenv.2022.114377.

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